## Applications

**1**

**a** $f(t) = 250 + 100\sin 2\pi t$

**b** $g(t) = 350 - 100\sin 4\pi t$

**c** 36 times

**d** $\frac{1}{8}$ years

**e** $t = \frac{7}{12}$

**2**

**a** $f\,(t) = 150 + 40\sin 2\pi t$

**b** $g\,(t) = 100 - 40\sin 3\pi t$

**c** 27 times

**d** $t = 1.70$

**3**

**a** Sine, the graph's initial value is at the midline.

**b** $I = \sin\left(\frac{\pi t}{2}\right)$

**c** 210 times

**4**

**a** 6 seconds

**b** $2500\ \text{mL}$

**c** $200\ \text{mL}$

**d** $60^\circ$ per second

**e** $c = \frac{1}{2}$

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**5**

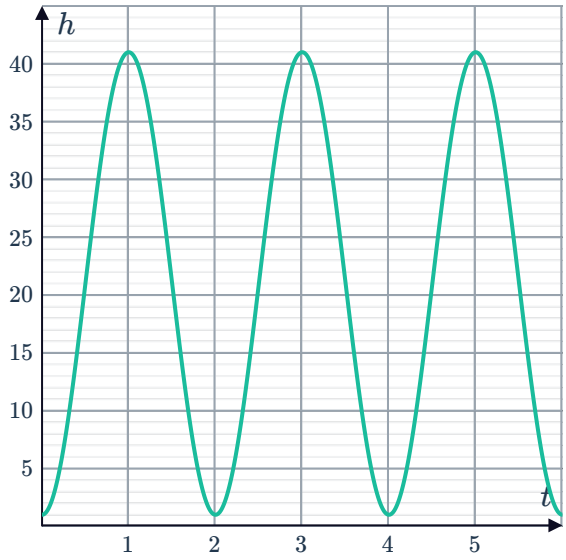
**a**

**i** No **ii** No **iii** No **iv** Yes

**b**

**i** 2 minutes **ii** $1\text{ m}$ **iii** $41\text{ m}$

**c**



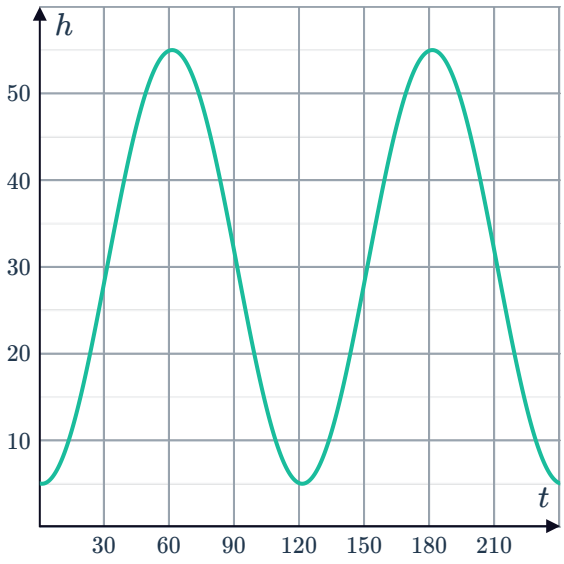

**d** $h = -20\cos(\pi t) + 21$

**e** $t = \dfrac{1}{2}$ minutes

**f** The height of the passengers above or below the centre of the wheel at time $t$.

**6**

**a**




**b** $55\text{ m}$

**c** $5\text{ m}$

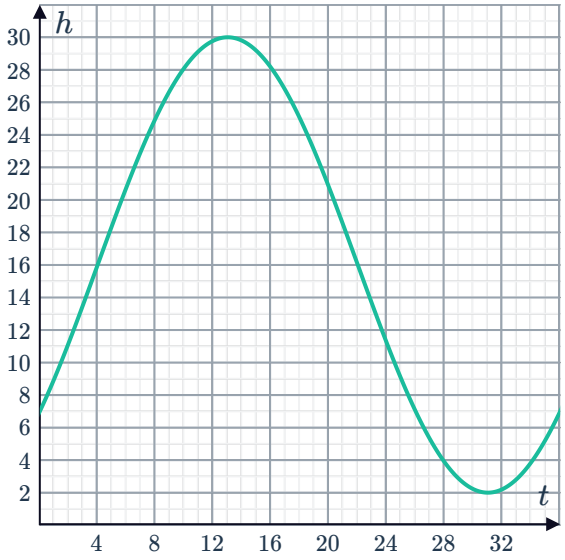
**d** $36.47\text{ m}$

**e** 120 seconds

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**7**

**a**




**b** $7.00\ \text{m}$

**c** $t = 34.82$ seconds

**d** $3$

**e** $3.88\ \text{m}$

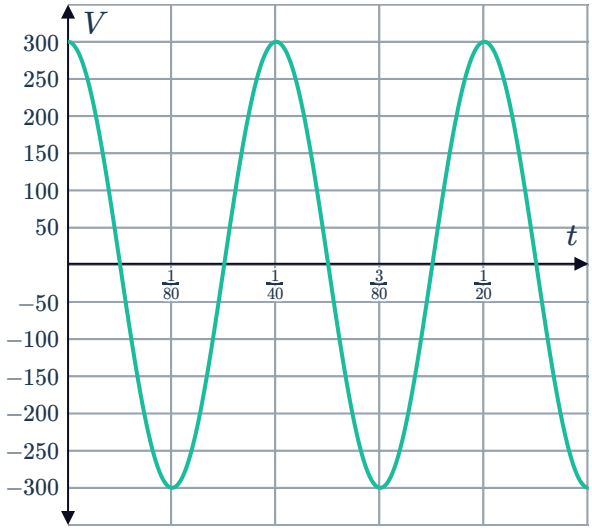
**8**

**a**

**i** Yes **ii** No **iii** No **iv** No

**b** $V = 300\cos(80\pi t)$

**c**




**d** It would decrease the period of the function and compress the graph horizontally.

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**9**

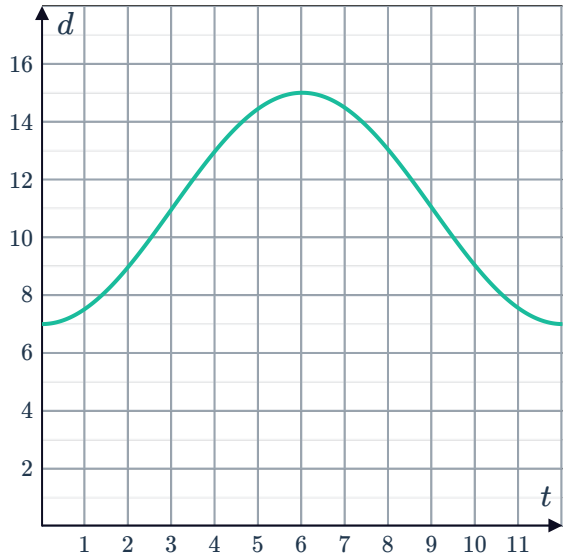
**a**

| Time ($t$, hours) | 0 | 3 | 6 | 9 | 12 |
|---|---|---|---|---|---|
| Tide level ($d$, metres) | 7 | 11 | 15 | 11 | 7 |

**b** 4 m

**c** 12 hours

**d**




**e**

**i** Yes **ii** No **iii** No **iv** No

**f** $d = -4\cos\frac{\pi}{6}t + 11$

**g** 2.66 hours

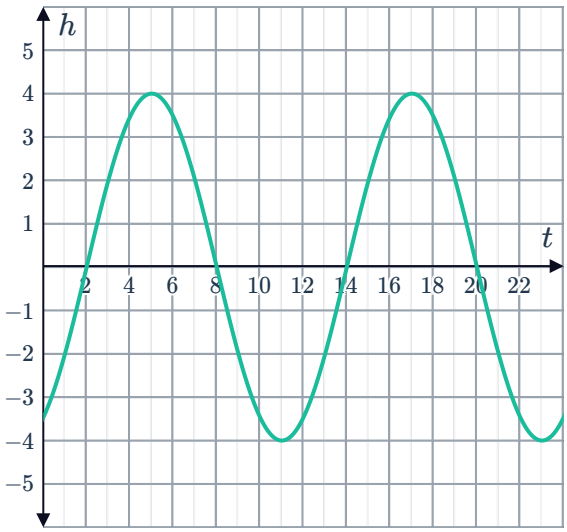
**10**

**a** $-2$ m

**b** 5:00 am

**c** 6 hours

**d**



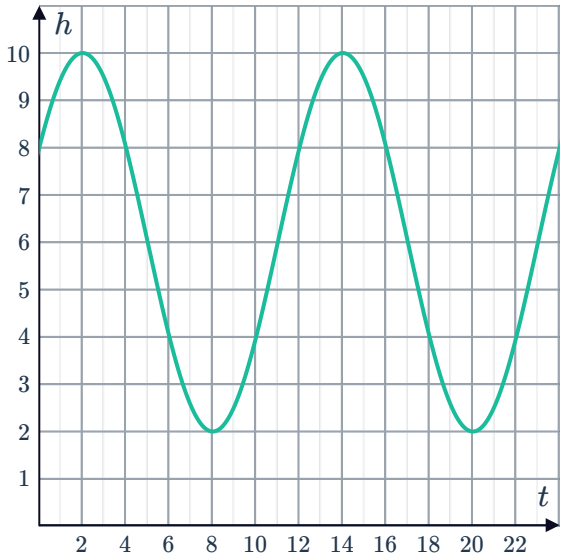

**e** 0 m

**f** 0.14 m

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**a** $h(t) = 6 + 4\cos\left(\frac{\pi}{6}(t-2)\right)$

**b**




**c** 2:00 am

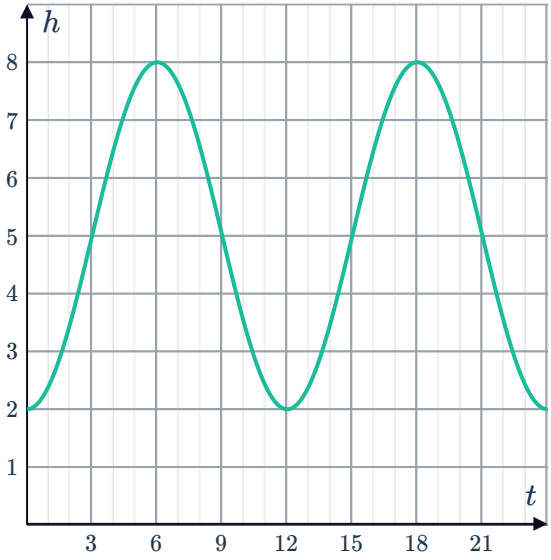
**d** 8:00 am

**12**

**a**

**i** $d = 5$ m **ii** $a = 3$ m **iii** $b = 30^\circ$/h **iv** $c = 90^\circ$

**b**




**c** 6 and 18 hours after noon

**13**

**a**

**i** $a = 40$ cm **ii** $d = 10$ cm **iii** $c = \frac{1}{4}$

**b** $t = \frac{1}{3}$ seconds

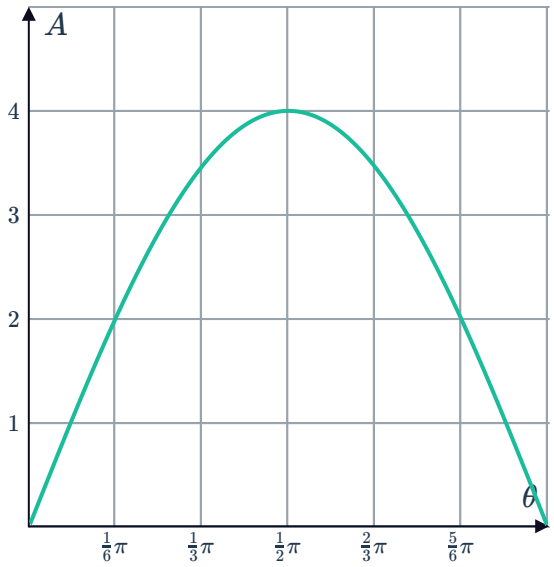
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**14**

**a** $A = 4\sin\theta$

**b** $0 < \theta < \pi$

**c**




**d** $\frac{\pi}{6}$

**15**

**a** 5 cm to the right of the centre

**b** To the left

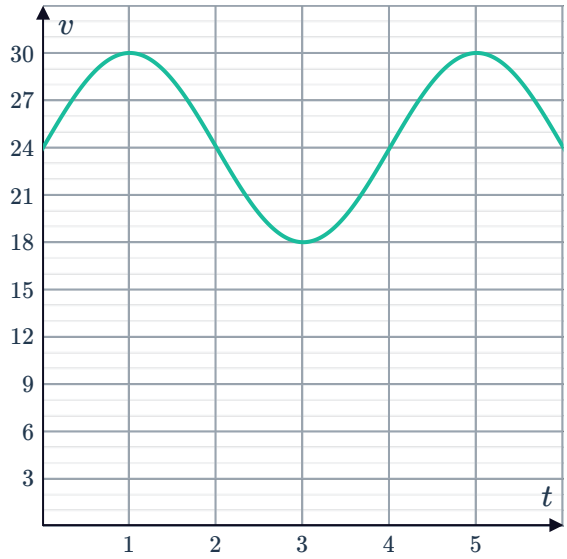
**c** 1 second

**d** 4 seconds

**e** $x = 5\cos\left(\frac{\pi t}{2}\right)$

**16**

**a**




**b** 30 units

**c** The number of seconds that have passed since the beginning of the section.

**d** 24 units

**17**

**a** 1.44 seconds

**b** 12.46 hours

**c** 110 days

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